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CANDIDATE NUMBER

SYDNEY GRAMMAR SCHOOL



2015 Trial Examination

FORM VI

MATHEMATICS EXTENSION II

Friday 31st July 2015

General Instructions

- Reading time — 5 minutes
- Writing time — 3 hour
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 100 Marks

- All questions may be attempted.

Section I – 10 Marks

- Questions 1–10 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II – 90 Marks

- Questions 11–16 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your candidate number on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Eleven.

Checklist

- SGS booklets — 6 per boy
- Multiple choice answer sheet
- Candidature — 73 boys

Examiner

BDD

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

The roots of the quadratic equation $x^2 - 8ix - 20 = 0$ are:

- | | |
|-----------------|-----------------|
| (A) $4i \pm 2$ | (B) $4 \pm 2i$ |
| (C) $-4i \pm 2$ | (D) $-4 \pm 2i$ |

QUESTION TWO

The value of $\int_0^{\frac{\pi}{2}} \sin x \cos x \, dx$ is:

- | | |
|--------------------|-------------------|
| (A) $-\frac{1}{2}$ | (B) $\frac{1}{4}$ |
| (C) $\frac{1}{2}$ | (D) 1 |

QUESTION THREE

The gradient of the tangent to the parametric curve $x = 2 \sec \theta$, $y = 3 \tan \theta$ at θ is:

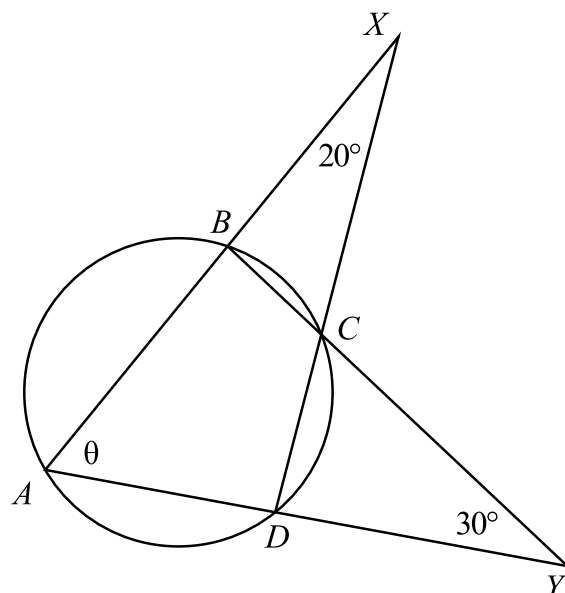
- | | |
|-------------------------------|---|
| (A) $\frac{2}{3} \sin \theta$ | (B) $\frac{2}{3} \operatorname{cosec} \theta$ |
| (C) $\frac{3}{2} \sin \theta$ | (D) $\frac{3}{2} \operatorname{cosec} \theta$ |

QUESTION FOUR

Which of the following functions is odd?

- | | |
|--------------------|-------------------------|
| (A) $y = x \sin x$ | (B) $y = \sin(\sin(x))$ |
| (C) $y = \ln x $ | (D) $y = \sin^2(x)$ |

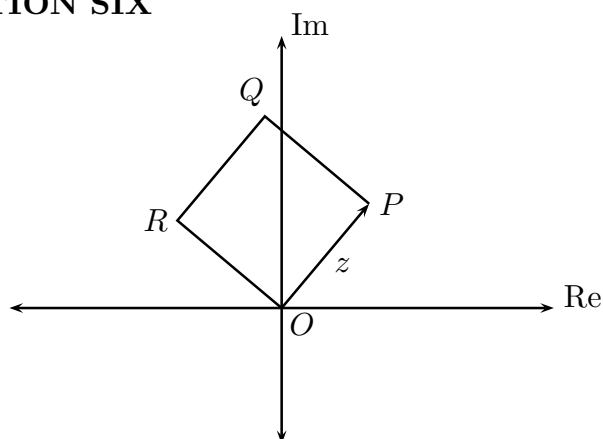
QUESTION FIVE



The size of angle θ in the diagram above is:

- | | |
|----------------|----------------|
| (A) 50° | (B) 55° |
| (C) 60° | (D) 65° |

QUESTION SIX



The point P in quadrant one represents complex number z . The points O , P , Q , R are the vertices of a square, as in the diagram.

Which statement is NOT true about the square:

- (A) side OR is represented by iz
- (B) the centre of the square is represented by $\frac{1}{2}(1 - i)z$
- (C) diagonal RP is represented by $(1 - i)z$
- (D) vertex Q is represented by $(1 + i)z$

Exam continues overleaf ...

QUESTION SEVEN

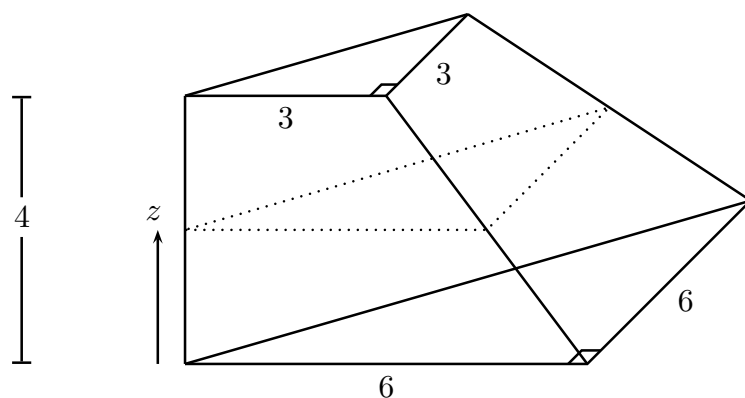
A pupil makes the following claims about the roots of the equation $z^6 = 1$:

- (I) The roots lie on the vertices of a hexagon in the complex plane
- (II) The roots lie on the unit circle in the complex plane
- (III) If ω is a root, then so is $\frac{1}{\omega}$
- (IV) If ω is a root, then so is $\bar{\omega}$

Which of these statements are TRUE?

- (A) I and IV
- (B) II and III
- (C) I, II and III
- (D) I, II, III and IV

QUESTION EIGHT



The base and top of the solid depicted are right angled isosceles triangles. A pupil is required to determine the volume by slicing parallel to the base. A typical slice parallel to the base at height z from the base is marked.

The cross-sectional area of the slice is:

- (A) $\frac{1}{2}(6 - \frac{3}{4}z)^2$
- (B) $\frac{1}{4}(6 - \frac{1}{4}z)^2$
- (C) $\frac{1}{2}(7 - z)^2$
- (D) $\frac{1}{2}(36 - \frac{27}{4}z)$

QUESTION NINE

The point defined by the complex number z moves in the complex plane subject to the constraint $|z - 3i| + |z + 3i| = 12$.

The locus of z is a conic with eccentricity:

(A) $\frac{1}{4}$.

(B) $\frac{1}{2}$.

(C) 1.

(D) 2.

QUESTION TEN

A polynomial $P(x)$ of fourth degree with real coefficients has the following properties:

$$P(1) = 0, P'(1) \neq 0$$

$$P(2) \neq 0, P'(2) = P''(2) = 0$$

What is the greatest number of complex non-real roots the polynomial could have?

(A) 0

(B) 1

(C) 2

(D) 3

————— End of Section I —————

Exam continues overleaf ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION ELEVEN (15 marks) Use a separate writing booklet.**Marks**

(a) Given $w = 1 - 2i$, $z = 3 + 4i$, express the following in the form $a + ib$ for real a , b :

(i) w^2 1

(ii) $\frac{z}{w}$ 1

(b) Given $t = 1 + i\sqrt{3}$ find:

(i) t in modulus–argument form, 2

(ii) t^8 in Cartesian form. 1

(c) Find:

(i) $\int \frac{1}{x^2 + 6x + 13} dx$ 2

(ii) $\int x \sin x dx$ 2

(d) Evaluate the following integral, expressing your answer in simplest exact form: 2

$$\int_{10}^{17} \frac{dx}{\sqrt{x^2 - 64}}.$$

(e) (i) Find constants A , B and C such that 2

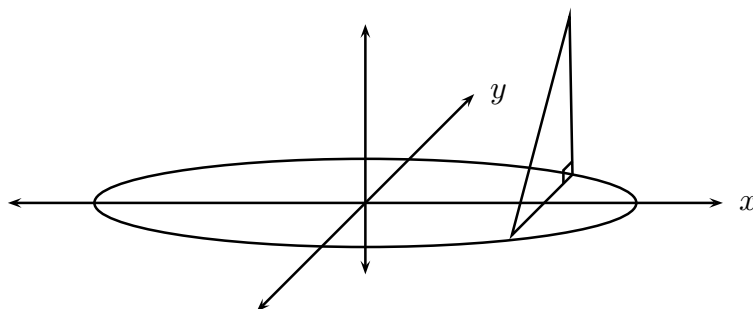
$$\frac{-4x^2 + 5x + 1}{(x - 1)^2} = \frac{A}{(x - 1)^2} + \frac{B}{x - 1} + C.$$

(ii) Hence find 2

$$\int \frac{-4x^2 + 5x + 1}{(x - 1)^2} dx.$$

QUESTION TWELVE (15 marks) Use a separate writing booklet.**Marks**(a) Consider the hyperbola $9x^2 - 16y^2 = 144$.(i) Find the eccentricity, foci, directrices and asymptotes of the hyperbola. 3(ii) Sketch the curve, locating the foci, directrices, y -intercepts and asymptotes. 2(iii) Use calculus to find the gradient of the tangent at $x = 5$ in quadrant one. 2(iv) Consider a tangent with point of contact in quadrant one. Explain geometrically why its gradient will always be greater than 0.75. 1(b) (i) Solve the equation $z^5 = -1$, leaving your answers in modulus-argument form. 1(ii) Hence factorise $z^5 + 1$ as a product of real linear and quadratic factors. 2

(c)



A certain solid has a base which is the ellipse $\frac{x^2}{16} + \frac{y^2}{4} = 1$. Slices perpendicular to the base and parallel to the y -axis are right-angled triangles of height 3 units.

(i) Show that the cross-sectional area of a slice parallel to the y -axis is 2

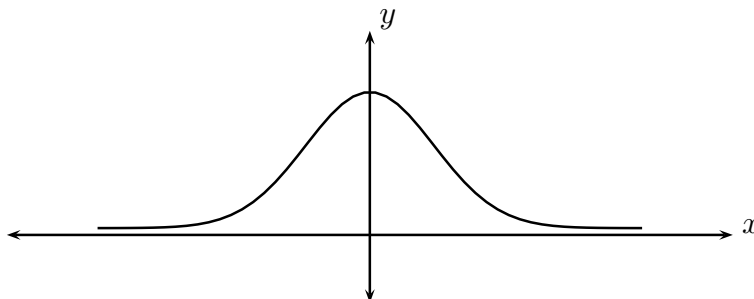
$$A(x) = \frac{3}{2} \sqrt{16 - x^2}$$

(ii) Hence find the volume of the solid. 2

QUESTION THIRTEEN (15 marks) Use a separate writing booklet.**Marks**

- (a) The cubic polynomial $P(x) = 2x^3 + 15x^2 + 24x + d$ is known to have a repeated real root and a distinct real root. The distinct root and repeated roots have opposite sign. Find the constant d . 3

(b)



The curve $y = e^{-x^2}$ is shown above.

- (i) Use the method of cylindrical shells to find the volume obtained when the region bounded by the axes, the curve and the line $x = 2$ is rotated about the y -axis. 3
- (ii) What is the limiting value as $N \rightarrow \infty$ of the volume obtained when the region bounded by the axes, the curve and the line $x = N$ is rotated about the y -axis? 1
- (c) A landing aeroplane of mass m kg is brought to rest by the action of two retarding forces: a force of $4m$ Newtons due to the reverse thrust of the engines; and a force due to the brakes of $\frac{mv^2}{40\,000}$ Newtons.

- (i) Show that the aeroplane's equation of motion for its speed v at time t seconds after landing is 1

$$\dot{v} = -\frac{v^2 + 400^2}{40\,000}.$$

- (ii) Assuming the aeroplane lands at a speed of U m/s, find an expression for the time it takes to come to rest. 3
- (iii) Show that, given a sufficiently long runway, then no matter how fast its landing speed, it will always come to rest within approximately 2.6 minutes of landing. 1

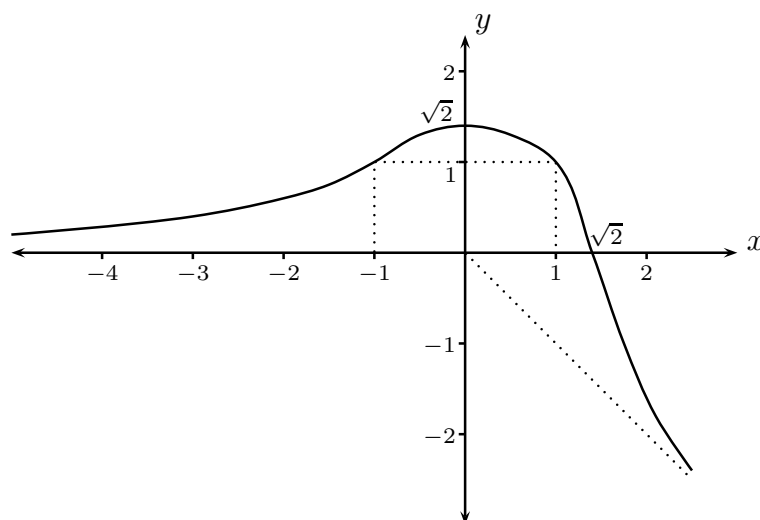
- (d) Consider the locus of z such that $|z - \sqrt{2} - i| = 1$.

- (i) Sketch the locus of z in the complex plane. 1
- (ii) Find the minimum value of $|z|$. 1
- (iii) Find the maximum value of $\arg(z)$, for $0^\circ < \arg(z) < 90^\circ$, correct to the nearest degree. 1

Exam continues next page ...

QUESTION FOURTEEN (15 marks) Use a separate writing booklet.**Marks**

(a)



The curve $y = f(x)$, sketched above, has asymptotes $y = 0$ and $y = -x$.

Copy or trace the above graph onto three separate number planes. Use your diagrams to show sketches of the following graphs, showing all essential features clearly.

(i) $y = (f(x))^2$

2

(ii) $|y| = f(x)$

2

(iii) $y = \ln f(x)$

2(b) (i) Let $z = \text{cis } \theta$. Use de Moivre's Theorem to prove that for any integer n ,**1**

$$z^n - \frac{1}{z^n} = 2i \sin n\theta.$$

(ii) By considering $\left(z + \frac{1}{z}\right)^5$, show that $\sin^5 \theta = \frac{1}{16}(\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta)$.**2**(iii) Solve the following equation for $0 \leq \theta \leq 2\pi$:**2**

$$\sin 5\theta - 5 \sin 3\theta + 9 \sin \theta = 0.$$

(c) You may assume the equation for the chord of contact for the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

from (x_0, y_0) is $\frac{x_0 x}{a^2} - \frac{y_0 y}{b^2} = 1$.

Show that that chord of contact from a point on a directrix is a focal chord.

2(d) Use the substitution $x = \frac{\pi}{2} - u$ to evaluate**2**

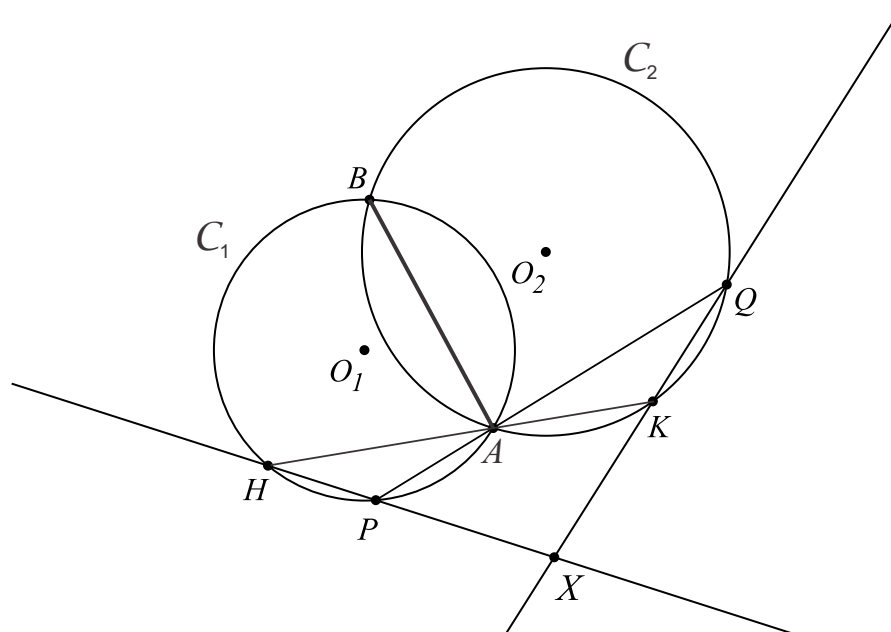
$$\int_0^{\frac{\pi}{2}} \frac{\cos^3 x}{\cos^3 x + \sin^3 x} dx.$$

Exam continues overleaf ...

QUESTION FIFTEEN (15 marks) Use a separate writing booklet.

Marks

(a)



Two intersecting circles C_1 and C_2 share a common chord AB . Points P and H lie on circle C_1 and points Q and K lie on circle C_2 , such that PAQ and HAK are straight. Lines HP and QK intersect at X .

Let $\angle BKQ = \theta$.

Copy or trace the diagram into your answer book.

- | | |
|---|---|
| (i) Find $\angle BAQ$ in terms of θ , giving a reason for your answer. | 1 |
| (ii) Show that $BKXH$ is cyclic. | 2 |
| (iii) Assuming that XAB is straight, show that XAB bisects angle PBK . | 2 |
- (b) (i) List all 10 ways that 3 non-negative integers can add to 3. 1
- (ii) Use the identity $(1+x)^{3n} = ((1+x)^n)^3$ to prove that 2
- $${}^{3n}C_3 = n^3 + 6n \times {}^nC_2 + 3 \times {}^nC_3$$

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QUESTION FIFTEEN (Continued)

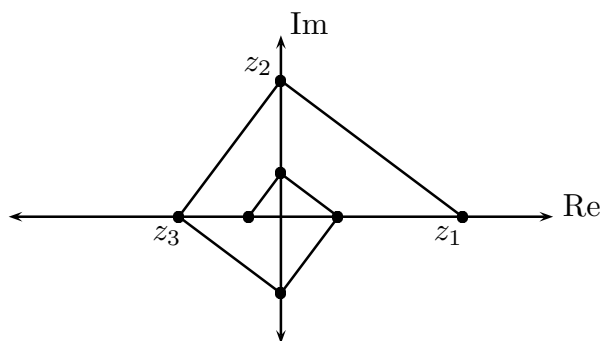
- (c) As part of a test of a new capsule delivery system, a capsule of mass m is fired straight up at speed u m/s. Air resistance is negligible and the magnitude of the acceleration due to gravity is g .

The capsule subsequently deploys a parachute and falls back to earth, subject to gravity and to a resistive force of magnitude mkv^2 .

- (i) Use calculus to show that the maximum height attained by the capsule is $H = \frac{u^2}{2g}$. **2**
- (ii) For the return trip, take the origin at the point it begins falling and assume down is positive. Show that the motion is determined by the equation $\ddot{x} = k(\alpha^2 - v^2)$, where $\alpha^2 = \frac{g}{k}$. **1**
- (iii) Let U be the impact speed of the package. Find an expression for the square of the speed U in terms of H , k and α . **3**
- (iv) Assume that the package is launched at speed $u = \alpha$. Find the impact speed as a percentage of the launch speed. **1**

QUESTION SIXTEEN (15 marks) Use a separate writing booklet.**Marks**

(a)



An infinite sequence of complex numbers is defined by

$$z_1 = 1, z_{n+1} = \frac{3}{4}iz_n.$$

The path $z_1z_2z_3\cdots$ defines a piecewise linear spiral in the Argand plane.

- (i) Show that the n^{th} edge satisfies the relationship **1**

$$z_{n+1} - z_n = \left(\frac{3}{4}i - 1\right) z_n.$$

- (ii) Hence find a simplified expression for the length of the n^{th} edge. **1**
- (iii) Find the length of the spiral, by considering the limiting sum of the lengths of its edges as $n \rightarrow \infty$. **1**

Exam continues overleaf ...

- (b) (i) Use algebra to prove, for any integer $k \geq 0$, that

1

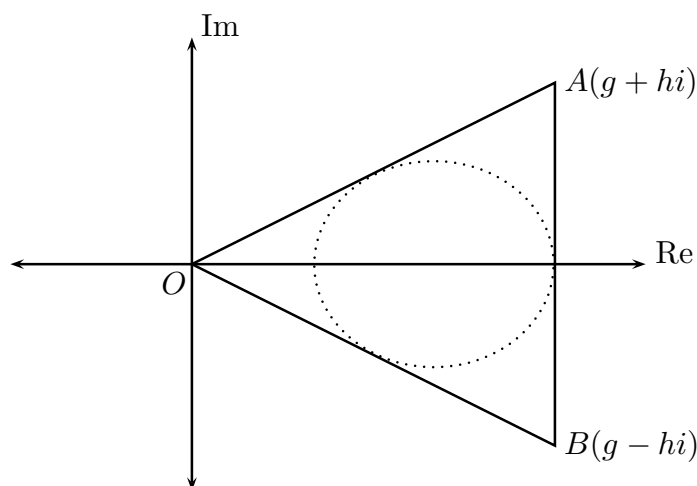
$$\frac{2k+1}{2k+2} \leq \frac{\sqrt{2k+1}}{\sqrt{2k+3}}.$$

- (ii) Prove, by induction on $n \geq 0$, that the central binomial coefficient $\binom{2n}{n}$ satisfies

3

$$\binom{2n}{n} \leq \frac{4^n}{\sqrt{2n+1}}.$$

(c)



Consider a cubic polynomial $y = P(x)$ with real coefficients and roots $0, g \pm hi$ where g and h are real and $h > 0$. In the diagram above, the roots form the vertices of an isosceles triangle OAB in the complex plane. The roots of $P'(x) = 0$ are the foci of the sketched ellipse which touches the triangle at $g + 0i$. We have sketched the case $g > 0$ and with major axis lying on the real axis. The centre of the ellipse is NOT the origin.

- (i) Show that the cubic polynomial has equation $y = x^3 - 2gx^2 + (g^2 + h^2)x$.

1

- (ii) Show that the turning points of $y = P(x)$, and hence the foci of the ellipse, occur at

1

$$x = \frac{2}{3}g \pm \frac{1}{3}\sqrt{g^2 - 3h^2}.$$

- (iii) Find the condition on g and h and hence on $\angle AOB$ which ensures that the major axis of this ellipse lies on the real axis. You may assume this condition holds in parts (iv) and (v).

1

- (iv) Find the equation of the ellipse.

2

- (v) Show that the ellipse is tangential to the triangle at the midpoints of OA and OB .

2

- (vi) If the triangle is equilateral, describe the behaviour of the polynomial $P(x)$ at the centre of the ellipse, for real x .

1

End of Section II

END OF EXAMINATION

SECTION I - Multiple Choice**QUESTION ONE**

The discriminant $\Delta = (-8i)^2 - 4 \times 1 \times (-20) = 16 = (4)^2$. Hence the roots are

$$\frac{8i \pm 4}{2} = 4i \pm 2$$

Hence **A**.

QUESTION TWO

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin x \cos x \, dx &= \frac{1}{2} \int_0^{\frac{\pi}{2}} \sin 2x \, dx \\ &= \left[-\frac{1}{4} \cos 2x \right]_0^{\frac{\pi}{2}} \\ &= \frac{1}{4} ((-\cos \pi) + \cos 0) \\ &= \frac{1}{2}\end{aligned}$$

Hence **C**.

QUESTION THREE

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy/d\theta}{dx/d\theta} \\ &= 3 \sec^2 \theta \div 2 \sec \theta \tan \theta \\ &= \frac{3}{2} \sec \theta \div \tan \theta \\ &= \frac{3}{2} \frac{1}{\cos \theta} \frac{\cos \theta}{\sin \theta} \\ &= \frac{3}{2} \operatorname{cosec} \theta\end{aligned}$$

Hence **D**.

QUESTION FOUR

Options A, C and D are even. Only option B is odd. Hence **B**.

QUESTION FIVE

Angle $XDY = \theta + 20$ (Exterior opposite angle in $\triangle AXD$)

Angle $BCD = \theta + 20 + 30$ (Exterior opposite angle in $\triangle DCY$)

Thus $\theta + (\theta + 50) = 180$ opposite angles of cyclic quad $ABCD$)

So $\theta = 65$.

Hence **D**.

QUESTION SIX

Option **C** is incorrect.

QUESTION SEVEN

All statements are TRUE. Hence **D**.

QUESTION EIGHT

The linear equation $y = (6 - \frac{3}{4}z)$ satisfies the conditions $y = 6$ when $z = 0$ and $y = 3$ when $z = 4$. The area of the triangle is $\frac{1}{2}y^2 = \frac{1}{2}(6 - \frac{3}{4})^2$, hence the correct answer is **A**.

QUESTION NINE

The foci are $\pm 3i$, hence the distance between the two foci is $2ae = 6$. But the sum of the distances from a point on the ellipse to the foci is $2a = 12$. Combining these two equations, $e = \frac{1}{2}$.

Hence the correct answer is **B**.

QUESTION TEN

The correct answer is 2. Note that there must be an even number of complex non-real roots, because of the real coefficients, and two is a possible answer. This is easily seen by drawing a polynomial with zero when $x = 1$, stationary point of inflexion when $x = 2$ and a turning point at some larger x value.

SECTION II - Written Response**QUESTION ELEVEN**

(a) (i) $w^2 = (1 - 2i)^2$

$$= 1 - 4 - 4i$$

$$= -3 - 4i$$

(ii) $\frac{z}{w} = \frac{3 + 4i}{1 - 2i}$

$$= \frac{(3 + 4i)(1 + 2i)}{1 + 4}$$

$$= \frac{3 - 8 + 4i + 6i}{5}$$

$$= -1 + 2i$$

(b) (i) We have $|t| = 2$, $\arg(t) = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$.

Hence $t = 2 \operatorname{cis} \frac{\pi}{3}$.

(ii) $t^8 = 2^8 \operatorname{cis} \frac{8\pi}{3}$

$$= 256 \operatorname{cis} \frac{2\pi}{3}$$

$$= 256\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)$$

$$= 128(-1 + i\sqrt{3})$$

(c) (i) $\int \frac{1}{x^2 + 6x + 13} dx = \int \frac{1}{(x + 3)^2 + 2^2} dx$

$$= \frac{1}{2} \tan^{-1} \frac{x + 3}{2} + C$$

(ii) $\int x \sin x dx = x(-\cos x) - \int 1 \times (-\cos x) dx$

$$= -x \cos x + \sin x + C$$

(d) $\int_{10}^{17} \frac{dx}{\sqrt{x^2 - 64}} = \left[\ln(x + \sqrt{x^2 - 64}) \right]_{10}^{17}$

$$= \ln(17 + \sqrt{17^2 - 64}) - \ln(10 + \sqrt{10^2 - 64})$$

$$= \ln 32 - \ln 16$$

$$= \ln 2$$

(e)

(i) $\frac{-4x^2 + 5x + 1}{(x - 1)^2} = \frac{A}{(x - 1)^2} + \frac{B}{x - 1} + C$

$$-4x^2 + 5x + 1 = A + B(x - 1) + C(x - 1)^2$$

Equating coefficients of x^2 tells us $C = -4$.
Substituting $x = 1$ tells us $A = 2$.

Substituting $x = 0$, tells us:

$$A - B + C = 1$$

$$2 - B + 4 = 1$$

$$B = -3$$

$$\begin{aligned} \text{(ii) Hence } \int \frac{-4x^2 + 5x + 1}{(x-1)^2} dx &= \int \frac{2}{(x-1)^2} dx + \int \frac{-3}{(x-1)} dx - \int 4 dx \\ &= \frac{-2}{(x-1)} - 3 \ln |x-1| - 4x + C \end{aligned}$$

QUESTION TWELVE

(a)

$$\text{(i) } \frac{x^2}{16} - \frac{y^2}{9} = 1$$

Hence $a = 4$ and $b = 3$.

$$\text{Now } b^2 = a^2(e^2 - 1)$$

$$e^2 = \frac{9}{16} + 1$$

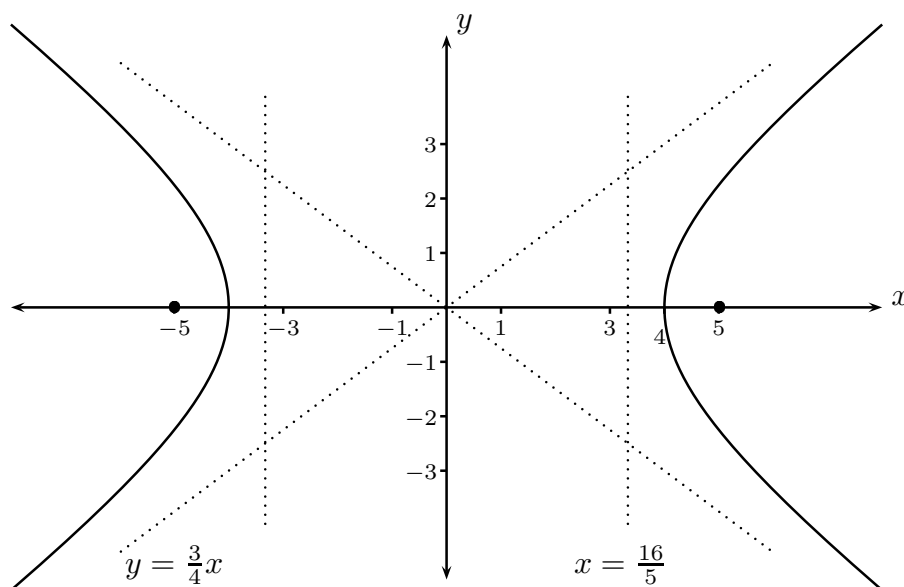
$$e^2 = \frac{25}{16}$$

$$e = \frac{5}{4}$$

The foci are $(\pm ae, 0) = (\pm 5, 0)$. The directrices are $x = \frac{a}{e}$ and $x = -\frac{a}{e}$. Thus $x = \frac{16}{5}$ and $x = -\frac{16}{5}$.

The asymptotes are $y = \frac{b}{a}x$ and $y = -\frac{b}{a}x$. Thus $y = \frac{3}{4}x$ and $y = -\frac{3}{4}x$.

(ii)



(iii) By substituting in the equation for the hyperbola, when $x = 5$, we find $y = \frac{9}{4}$ in quadrant one. Differentiating with respect to x :

$$\begin{aligned}
 18x - 32y \frac{dy}{dx} &= 0 \\
 \frac{dy}{dx} &= \frac{9x}{16y} \\
 &= \frac{45}{36} \quad \text{when } x = 5 \\
 &= 1.25
 \end{aligned}$$

(iv) The tangent in Quadrant One will be steeper than the asymptote, thus its gradient will never be less than that of the asymptote.

(b) (i) Let $z = \text{cis } \theta$. Then

$$z^5 = -1$$

$$\text{cis } 5\theta = \text{cis}(\pi + 2k\pi) \quad \text{for any integer } k$$

Equating arguments gives:

$$5\theta = (2k + 1)\pi$$

$$\theta = \frac{\pi}{5}, \frac{3\pi}{5}, \pi, -\frac{\pi}{5}, -\frac{3\pi}{5}$$

Hence (in conjugate pairs) the roots are:

$$z = -1, \quad z = \text{cis}\left(\pm\frac{1}{5}\pi\right), \quad z = \text{cis}\left(\pm\frac{3}{5}\pi\right)$$

(ii) Grouping the conjugate roots, we get:

$$\begin{aligned}
 z^5 + 1 &= (z + 1) \times \left(z - \text{cis}\frac{1}{5}\pi\right) \left(z - \text{cis}\left(-\frac{1}{5}\pi\right)\right) \times \left(z - \text{cis}\frac{3}{5}\pi\right) \left(z - \text{cis}\left(-\frac{3}{5}\pi\right)\right) \\
 &= (z + 1) \left(z^2 - 2\cos\left(\frac{1}{5}\pi\right)z + 1\right) \left(z^2 - 2\cos\left(\frac{3}{5}\pi\right)z + 1\right)
 \end{aligned}$$

(c) (i) The area of the triangle is:

$$\begin{aligned}
 \frac{1}{2}bh &= \frac{1}{2}(2y) \times 3 \\
 &= 3y \\
 &= 6 \times \sqrt{1 - \frac{x^2}{16}} \\
 &= \frac{6}{4} \times \sqrt{16 - x^2} \\
 &= \frac{3}{2} \sqrt{16 - x^2}
 \end{aligned}$$

(ii) The volume is

$$\begin{aligned}
 V &= 3 \times \int_0^4 \sqrt{16 - x^2} \, dx \\
 &= 3 \times \frac{1}{4} \pi 4^2
 \end{aligned}$$

since the integral is the area of a quarter circle of radius 4.

Thus the volume is $V = 12\pi$.

(This integral can also be evaluated using the trig substitution $x = 4 \sin u$.)

QUESTION THIRTEEN

- (a) At a double root we have a zero of the derivative.

$$P'(x) = 6x^2 + 30x + 24$$

$$= 6(x + 4)(x + 1)$$

Hence the possibilities are $x = -1$ or $x = -4$. The other root must be positive and the product of roots must be negative, i.e. $d < 0$.

If $x = -1$ then $P(-1) = 2(-1)^3 + 15(-1)^2 + 24(-1) + d$

$$0 = -2 + 15 - 24 + d$$

$$d = 11$$

If $x = -4$ then $P(-4) = 2(-4)^3 + 15(-4)^2 + 24(-4) + d$

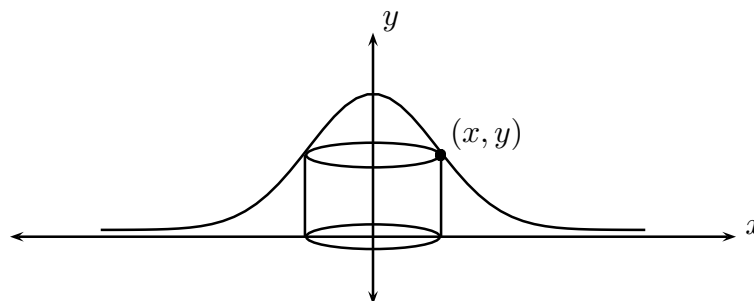
$$0 = -128 + 240 - 96 + d$$

$$d = -16$$

Hence $d = -16$.

This question can also be solved using sum and product of roots methods.

- (b) (i)



The volume of the cylindrical shell $dV = 2\pi xy dx$. Total volume is

$$\begin{aligned} V &= \int_0^2 2\pi xy \, dx \\ &= \pi \int_0^2 2xe^{-x^2} \, dx \\ &= \pi \times [e^{-x^2}]_0^2 \\ &= \pi \times (1 - e^{-4}) \end{aligned}$$

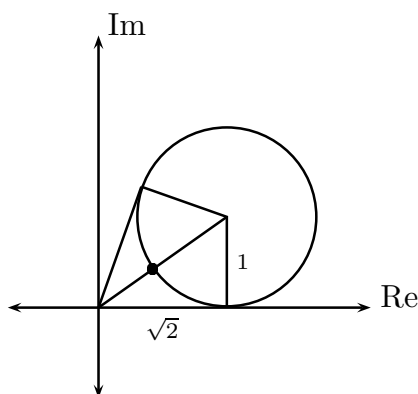
$$\begin{aligned} \text{(ii)} \quad V &= \pi \times \lim_{N \rightarrow \infty} (1 - e^{-N^2}) \\ &= \pi \end{aligned}$$

$$\begin{aligned} \text{(c) (i)} \quad m\dot{v} &= \frac{-mv^2}{40\,000} - 4m \\ \dot{v} &= -\frac{v^2}{40\,000} - 4 \\ \dot{v} &= -\frac{v^2 + 160\,000}{40\,000} \\ &= -\frac{v^2 + 400^2}{40\,000} \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \frac{dv}{dt} &= -\frac{v^2 + 400^2}{40\,000} \\
 dt &= \frac{-40\,000\,dv}{v^2 + 400^2} \\
 \int_0^T dt &= -40\,000 \times \int_U^0 \frac{dv}{v^2 + 400^2} \\
 T &= 40\,000 \times \frac{1}{400} \tan^{-1} \frac{U}{400} \\
 T &= 100 \tan^{-1} \frac{U}{400}
 \end{aligned}$$

(iii) As $U \rightarrow \infty$, $T \rightarrow 100 \times \frac{\pi}{2}$ seconds, which is about 2.6 minutes.

(d) (i)



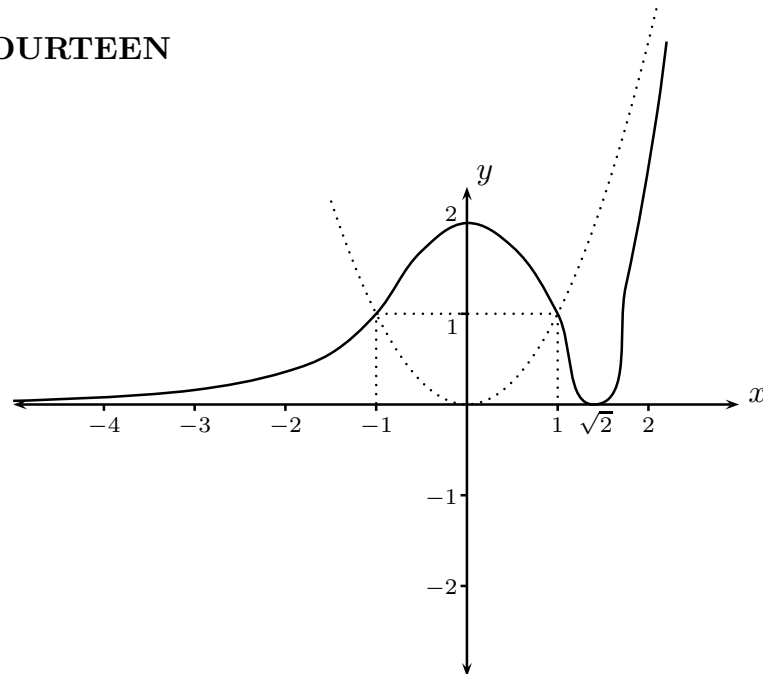
(ii) The point z of minimum modulus is the point on the circle closest to the origin. This distance is:

$$(\text{distance from origin to centre of circle}) - (\text{radius}) = \sqrt{3} - 1$$

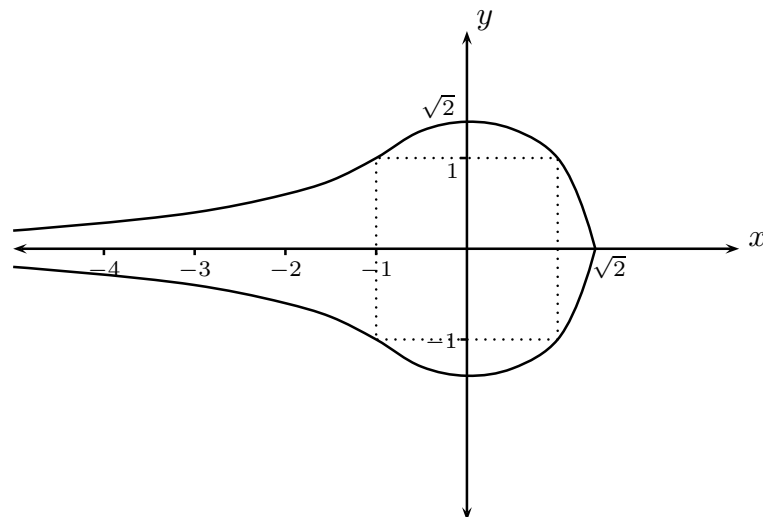
(iii) The point with maximum $\arg(z)$ on the circle is defined by the tangent to the circle. The argument is $2 \times \tan^{-1} \frac{1}{\sqrt{2}} \doteq 71^\circ$.

QUESTION FOURTEEN

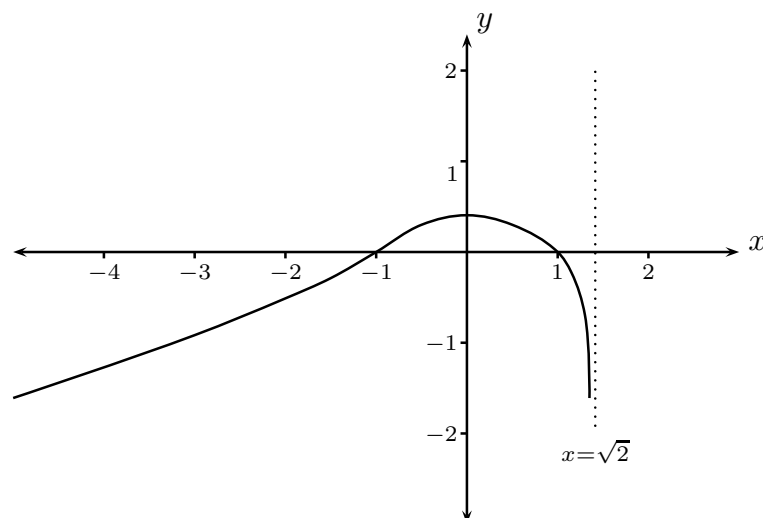
(a) (i)



(ii)



(iii)



(b) (i) Let $z = \text{cis } \theta$. Then by de Moivre's Theorem,

$$\begin{aligned} z^n - \frac{1}{z^n} &= (\text{cis } \theta)^n - (\text{cis } \theta)^{-n} \\ &= \text{cis}(n\theta) - \text{cis}(-n\theta) \\ &= \cos(n\theta) + i \sin(n\theta) - \cos(-n\theta) - i \sin(-n\theta) \\ &= \cos(n\theta) + i \sin(n\theta) - \cos(n\theta) + i \sin(n\theta) \\ &= 2i \sin(n\theta) \end{aligned}$$

Where we have used the evenness of the cosine function and the oddness of the sine function.

$$\begin{aligned} \text{(ii)} \quad \left(z - \frac{1}{z}\right)^5 &= z^5 - 5z^4 \frac{1}{z} + 10z^3 \frac{1}{z^2} - 10z^2 \frac{1}{z^3} + 5z \frac{1}{z^4} - \frac{1}{z^5} \\ &= \left(z^5 - \frac{1}{z^5}\right) - 5\left(z^3 - \frac{1}{z^3}\right) + 10\left(z - \frac{1}{z}\right) \\ &= 2i \sin 5\theta - 5 \times 2i \sin 3\theta + 10 \times 2i \sin \theta \end{aligned}$$

Now the LHS of this expression is $(2i \sin \theta)^5$, hence

$$32i \sin^5 \theta = 2i \sin 5\theta - 5 \times 2i \sin 3\theta + 10 \times 2i \sin \theta$$

$$16 \sin^5 \theta = \sin 5\theta - 5 \sin 3\theta + 10 \sin \theta$$

$$\sin^5 \theta = \frac{1}{16} \left(\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta \right)$$

(iii) From this equation $16 \sin^5 \theta - 10 \sin \theta = \sin 5\theta - 5 \sin 3\theta$.

$$\text{Hence} \quad \sin 5\theta - 5 \sin 3\theta + 9 \sin \theta = 0$$

$$\text{Becomes } 16 \sin^5 \theta - 10 \sin \theta + 9 \sin \theta = 0$$

$$16 \sin^5 \theta - \sin \theta = 0$$

$$\sin \theta (16 \sin^4 \theta - 1) = 0.$$

$$\text{So } \sin \theta = 0 \text{ or } \sin \theta = \pm \frac{1}{2}.$$

The solutions of these equations in the given domain are:

$$\theta = 0, \pi, 2\pi, \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

(c) Let the point be $(x_0, y_0) = \left(\frac{a}{e}, y_0\right)$. The chord of contact is $\frac{x}{a^2} \left(\frac{a}{e}\right) - \frac{y_0 y}{b^2} = 1$.

Is $(ae, 0)$ on this chord?

$$LHS = \frac{x}{a^2} \left(\frac{a}{e}\right) - \frac{y_0 y}{b^2}$$

$$= 1 - 0$$

$$= RHS$$

So yes, the chord passes through the focus.

$$\begin{aligned}
 \text{(d)} \quad \int_0^{\frac{\pi}{2}} \frac{\cos^3 x}{\cos^3 x + \sin^3 x} dx &= \int_{\frac{\pi}{2}}^0 \frac{\cos^3(\frac{\pi}{2} - u)}{\cos^3(\frac{\pi}{2} - u) + \sin^3(\frac{\pi}{2} - u)} (-du) \\
 &= \int_0^{\frac{\pi}{2}} \frac{\sin^3 u}{\sin^3 u + \cos^3 u} du \\
 &= \int_0^{\frac{\pi}{2}} \frac{\sin^3 x}{\sin^3 x + \cos^3 x} dx \quad (\text{relabelling } u \text{ as } x)
 \end{aligned}$$

$$\begin{aligned}
 \text{Hence } 2 \times \int_0^{\frac{\pi}{2}} \frac{\cos^3 x}{\cos^3 x + \sin^3 x} dx &= \int_0^{\frac{\pi}{2}} \frac{\cos^3 x}{\cos^3 x + \sin^3 x} dx + \int_0^{\frac{\pi}{2}} \frac{\sin^3 x}{\sin^3 x + \cos^3 x} dx \\
 &= \int_0^{\frac{\pi}{2}} \frac{\cos^3 x + \sin^3 x}{\cos^3 x + \sin^3 x} dx \\
 &= \int_0^{\frac{\pi}{2}} 1 dx \\
 &= \frac{\pi}{2}
 \end{aligned}$$

$$\text{Thus } \int_0^{\frac{\pi}{2}} \frac{\cos^3 x}{\cos^3 x + \sin^3 x} dx = \frac{\pi}{4}.$$

QUESTION FIFTEEN

- (a) (i) $\angle BAQ = \theta$ (angles at the circumference on arc BQ)
(ii) Hence $\angle BHP = \theta$ (interior opposite angle in cyclic quadrilateral $BHPA$)
Since $\angle BHP = \angle BHX = \theta$ and the exterior opposite angle $\angle BKQ = \theta$, we have that quadrilateral $BKXH$ is cyclic.
(iii) $\angle PBA = \angle PHA$ (angles at the circumference on arc PA in circle $\mathcal{C}_1$)
 $= \angle XHK$ (same angle)
 $= \angle XBK$ (angles at the circumference on arc XK in circle $BKXH$)
 $= \angle ABK$ (same angle)

- (b) (i) In any order, the ways three integers can add to 3 are:

$$\begin{aligned}
 &3 + 0 + 0, 0 + 3 + 0, 0 + 0 + 3 \\
 &1 + 2 + 0, 2 + 1 + 0, 0 + 2 + 1, 0 + 1 + 2, 1 + 0 + 2, 2 + 0 + 1 \\
 &1 + 1 + 1
 \end{aligned}$$

- (ii) We look to be equating coefficients of x^3 .

$$\begin{aligned}
 LHS &= (1 + x)^{3n} \\
 &= {}^{3n}C_0 x^0 + {}^{3n}C_1 x^1 + \dots
 \end{aligned}$$

... and the coefficient of x^3 is ${}^{3n}C_3$.

The RHS is $(1 + x)^n (1 + x)^n (1 + x)^n$.

We need to consider how we can get an x^3 term when we expand the brackets. Part (i) gives us a hint here, since the sum of the three indices (one from each bracket) must be 3.

Method 1 (list them all): The coefficient of x^3 is:

$$\begin{aligned} & {}^nC_3 {}^nC_0 {}^nC_0 + {}^nC_0 {}^nC_3 {}^nC_0 + {}^nC_0 {}^nC_0 {}^nC_3 \\ & + {}^nC_1 {}^nC_2 {}^nC_0 + {}^nC_2 {}^nC_1 {}^nC_0 + {}^nC_0 {}^nC_2 {}^nC_1 + {}^nC_0 {}^nC_1 {}^nC_2 + {}^nC_1 {}^nC_0 {}^nC_2 \\ & + {}^nC_2 {}^nC_0 {}^nC_1 \\ & + {}^nC_1 {}^nC_1 {}^nC_1 \\ & = 3 \times {}^nC_3 + 6 \times {}^nC_1 {}^nC_2 + ({}^nC_1)^3 \end{aligned}$$

Method 2 (Avoid listing them all):

- There are 3 ways to get x^3 from one bracket, x^0 from each of the others.
This gives a contribution $3 \times {}^nC_3 x^3 \times x^0 \times x^0$
- There are 6 ways to get x^1 from one bracket, x^2 from a second bracket and x^0 from a third.
This gives a contribution $6 \times {}^nC_1 x^1 \times {}^nC_2 x^2 \times x^0$
- There is 1 way to get x^1 from all of the brackets
This gives a contribution $1 \times {}^nC_1 x \times {}^nC_1 x \times {}^nC_1 x$

Thus from the *RHS* the coefficient of x^3 is:

$$3 \times {}^nC_3 + 6 \times {}^nC_1 {}^nC_2 + ({}^nC_1)^3$$

Thus either method gives us

$$\begin{aligned} & 3 \times {}^nC_3 + 6 \times {}^nC_1 {}^nC_2 + ({}^nC_1)^3 \\ & = 3 \times {}^nC_3 + 6n \times {}^nC_2 + n^3 \end{aligned}$$

Equating coefficients of x^3 yields the required result.

- (c) (i) Starting from the equation of motion $\ddot{x} = -g$, we get

$$\begin{aligned} \ddot{x} &= -g \\ v \frac{dv}{dx} &= -g \\ \int_u^0 v dv &= - \int_0^H g dx \\ \left[\frac{1}{2} v^2 \right]_u^0 &= \left[-gx \right]_0^H \\ -gH &= -\frac{1}{2} u^2 \\ H &= \frac{u^2}{2g} \end{aligned}$$

- (ii) Working the equations with down as positive and including the resistive term;

$$m\ddot{x} = mg - mkv^2$$

$$\ddot{x} = g - kv^2$$

$$= k \left(\frac{g}{k} - v^2 \right)$$

$$= k(\alpha^2 - v^2)$$

$$\text{where } \alpha^2 = \frac{g}{k}.$$

(iii) We need to integrate the equation of motion:

$$v \frac{dv}{dx} = k(\alpha^2 - v^2)$$

$$-\frac{1}{2} \times \int_0^U \frac{-2v dv}{\alpha^2 - v^2} = k \times \int_0^H dx$$

$$\left[-\frac{1}{2} \ln(\alpha^2 - v^2)\right]_0^U = [kx]_0^H$$

$$-\frac{1}{2} \ln \frac{\alpha^2 - U^2}{\alpha^2} = kH$$

Hence $kH = -\frac{1}{2} \ln \frac{\alpha^2 - U^2}{\alpha^2}$. (Note that $U < \alpha$, the terminal velocity.)

Rearranging we get:

$$U^2 = \alpha^2(1 - e^{-2kH})$$

(iv) We have:

$$\frac{(\text{impact speed})^2}{(\text{launch speed})^2} = \frac{\alpha^2(1 - e^{-2kH})}{\alpha^2}$$

$$= 1 - e^{-2kH}$$

But $2kH = 2 \times \frac{g}{\alpha^2} \times \frac{\alpha^2}{2g} = 1$. Hence

$$\frac{(\text{impact speed})^2}{(\text{launch speed})^2} = 1 - e^{-1}$$

$$= \frac{e - 1}{e}$$

$$\doteq 0.632$$

So $\frac{\text{impact speed}}{\text{launch speed}} = 79.5\%$

That is, the impact speed = 79.5% of the launch speed.

QUESTION SIXTEEN

(a) (i) $z_{n+1} - z_n = \frac{3}{4}iz_n - z_n$

$$= \left(\frac{3}{4}i - 1\right)z_n$$

(ii) $|z_{n+1} - z_n| = \left|\frac{3}{4}i - 1\right| \times |z_n|$

$$= \frac{5}{4}|z_n|$$

But $|z_n| = \left(\frac{3}{4}\right)^{n-1}$, so

(iii) $|z_{n+1} - z_n| = \frac{5}{4} \times \left(\frac{3}{4}\right)^{n-1}$

$$\text{Total length} = \frac{5}{4}\left(1 + \frac{3}{4} + \frac{9}{16} + \cdots\right)$$

$$= \frac{5}{4} \times \frac{1}{1 - \frac{3}{4}}$$

$$= \frac{5}{4} \times 4$$

$$= 5$$

(b) Consider RHS/LHS . We want to show that this ratio is greater than 1.

$$\begin{aligned}\left(\frac{RHS}{LHS}\right)^2 &= \frac{(2k+1)(2k+2)^2}{(2k+3)(2k+1)^2} \\ &= \frac{(2k+2)^2}{(2k+3)(2k+1)} \\ &= \frac{4k^2 + 8k + 4}{4k^2 + 8k + 3} \\ &> 1\end{aligned}$$

(Since the numerator is greater than the denominator.)

(i)

Step A: Let us check the result for $n = 0$. When $n = 0$:

$$\begin{aligned}LHS &= \binom{0}{0} & RHS &= \frac{4^n}{\sqrt{2n+1}} \\ &= 1 & &= 1\end{aligned}$$

Step B: Assume the results holds for $n = k$, that is assume

$$\binom{2k}{k} \leq \frac{4^k}{\sqrt{2k+1}}$$

We need to show that the result holds for $n = k + 1$, that is to show that

$$\binom{2k+2}{k+1} \leq \frac{4^{k+1}}{\sqrt{2k+3}}$$

$$\begin{aligned}LHS &= \frac{(2k+2)!}{(k+1)!(k+1)!} \\ &= \frac{(2k)!}{k!k!} \times \frac{(2k+1)(2k+2)}{(k+1)(k+1)} \\ &\leq \frac{4^k}{\sqrt{2k+1}} \times \frac{4(2k+1)}{(2k+2)} \\ &\leq \frac{4^{k+1}}{\sqrt{2k+1}} \times \frac{\sqrt{2k+1}}{\sqrt{2k+3}} \\ &\leq \frac{4^{k+1}}{\sqrt{2k+3}}\end{aligned}$$

as required.

Step C: Hence the result holds for all n by the Principle of Mathematical Induction.

(c) (i) $y = x(x - (g + hi))(x - (g - hi))$

$$= x(x^2 - 2gx + (g^2 + h^2))$$

$$= x^3 - 2gx^2 + x(g^2 + h^2)$$

(ii) $y' = 3x^2 - 4gx + (g^2 + h^2)$

So $y' = 0$ when

$$x = \frac{4g \pm \sqrt{16g^2 - 4 \times 3 \times (g^2 + h^2)}}{6}$$

$$= \frac{2}{3}g \pm \frac{1}{3}\sqrt{g^2 - 3h^2}$$

- (iii) We need $g^2 - 3h^2 \geq 0$, so that the foci are real.

$$\text{Thus } \frac{h^2}{g^2} \leq \frac{1}{3}$$

$$\frac{h}{g} \leq \frac{1}{\sqrt{3}}$$

But $\tan \frac{1}{2} \angle AOB = \frac{h}{g}$, so $\frac{1}{2} \angle AOB \leq 30^\circ$. Thus $\angle AOB \leq 60^\circ$.

- (iv) The centre of the ellipse occurs at the midpoint of the foci, which are the stationary points of $P(x)$. Thus the centre is at $x = \frac{2}{3}g$.

The equation of the ellipse is

$$\frac{(x - \frac{2}{3}g)^2}{a^2} + \frac{y^2}{b^2} = 1$$

The endpoint of the ellipse is given to be $(g, 0)$, so that $a = g - \frac{2}{3}g = \frac{1}{3}g$. The distance between the foci is $2ae$, so using part (i)

$$2ae = \frac{2}{3}\sqrt{g^2 - 3h^2}$$

$$\begin{aligned} \text{Thus } b^2 &= a^2(1 - e^2) \\ &= a^2 - (ae)^2 \\ &= \frac{1}{9}g^2 - \frac{1}{4} \times \frac{4}{9}(g^2 - 3h^2) \\ &= \frac{1}{3}h^2 \end{aligned}$$

Hence the equation is

$$\frac{(x - \frac{2}{3}g)^2}{\frac{1}{9}g^2} + \frac{y^2}{\frac{1}{3}h^2} = 1$$

- (v) We need to show that $(\frac{1}{2}g, \frac{1}{2}h)$ lies on the ellipse, and that at this point the ellipse has gradient $\frac{h}{g}$.

Substituting in the equation for the ellipse:

$$\begin{aligned} LHS &= \frac{(\frac{1}{2}g - \frac{2}{3}g)^2}{\frac{1}{9}g^2} + \frac{(\frac{1}{2}h)^2}{\frac{1}{3}h^2} \\ &= \frac{\frac{1}{36}g^2}{\frac{1}{9}g^2} + \frac{3}{4} \\ &= \frac{1}{4} + \frac{3}{4} \\ &= 1 \end{aligned}$$

Hence the point lies on the ellipse.

By implicit differentiation of the equation of the ellipse:

$$\frac{2(x - \frac{2}{3}g)}{\frac{1}{9}g^2} + \frac{2yy'}{\frac{1}{3}h^2} = 0$$

$$\begin{aligned}
\text{Thus } y' &= \frac{-2(x - \frac{2}{3}g)}{\frac{1}{9}g^2} \times \frac{\frac{1}{3}h^2}{2y} \\
&= \frac{-2(\frac{1}{2}g - \frac{2}{3}g)}{\frac{1}{9}g^2} \times \frac{\frac{1}{3}h^2}{2\frac{1}{2}h} \\
&= \frac{-2(-\frac{1}{6}g)}{\frac{1}{9}g^2} \times \frac{1}{3}h \\
&= \frac{3}{g} \times \frac{1}{3}h \\
&= \frac{h}{g}
\end{aligned}$$

Which is the gradient of OA , and hence the ellipse is tangential to the triangle at the midpoint of OA .

A similar proof (not required) would show that the ellipse is tangential to the triangle at the midpoint of OB also.

- (vi) Angle $AOB = 60^\circ$, hence $\frac{h}{g} = \tan 30^\circ = \frac{1}{\sqrt{3}}$. Thus $g^2 - 3h^2 = 0$. The foci are both $x = \frac{2}{3}g$ and the ellipse is a circle, with centre $(\frac{2}{3}g, 0)$ and radius $\frac{1}{3}g$.

Note: At $x = \frac{2}{3}g$ the polynomial has a double root of $P'(x)$ and a point of inflexion – it defines a stationary point of inflexion.